

Notes: Diamond, Jiang & Ma (JFE, 2024)
The reserve supply channel of unconventional monetary policy

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1 Big picture

- **Question.** What is the aggregate effect of central bank reserve injections created by quantitative easing (QE) on bank lending?
- **Main finding.** QE reserves crowd out bank lending. The paper estimates that the reserves injected by QE from 2008–2017 raised corporate loan rates by 7.37 basis points and reduced bank loans by about \$134.2 billion per year.
- **Reserve supply channel.** QE creates reserves that must be held by banks. Holding additional safe liquid assets consumes balance-sheet capacity and raises the marginal cost of providing loans and mortgages.
- **Key estimate.** Each dollar of reserves created by QE reduces bank lending by about 7.67 cents. This per-dollar number is small, but the aggregate amount of reserves is very large.
- **Why loans are the main margin.** Corporate loan demand is much more interest-rate elastic than deposit or mortgage demand. When banks must hold more reserves, loan quantities adjust more than deposits or mortgages.
- **Structural approach.** The paper estimates a model with deposit, mortgage, and loan demand systems plus a bank balance-sheet cost function. It then simulates counterfactual reserve injections.
- **Identification strategy.** Demand elasticities are estimated using cross-sectional instruments based on natural disasters and internal-capital-market exposure. Balance-sheet cost parameters are estimated using natural-disaster shocks, a Bartik deposit shock, and high-frequency changes in the Treasury General Account (TGA).
- **Main mechanism.** Additional securities/reserves raise the marginal cost of lending and mortgages but lower the marginal cost of providing deposits. This pushes loan rates up and crowds out loans.
- **Model validation.** The model-implied reserve spread tracks the IOER–federal funds spread reasonably well, even though that time series is not targeted directly in the structural counterfactual.
- **Policy implication.** QE’s asset-purchase channel may stimulate the economy, but the associated reserve supply can reduce bank loan supply. Reducing banks’ cost of holding reserves, for example by exempting reserves from leverage constraints, could weaken the crowding-out effect.

2 Data and measurement

2.1 What the paper measures

The empirical object is the banking system's response to a reserve supply shock. The paper measures how a forced increase in banks' holdings of reserves and other liquid securities affects:

- deposit rates and deposit quantities;
- mortgage rates and mortgage quantities;
- corporate loan rates and corporate loan quantities;
- bank securities and reserve holdings.

Interpretation. The paper does not ask whether QE lowers long-term yields through asset purchases. It isolates the separate channel created by the reserves used to pay for those purchases.

2.2 Bank products and balance-sheet quantities

Banks operate in local markets indexed by n , banks by m , and time by t . The key balance-sheet quantities are:

$$Q_{D,nmt}, \quad Q_{M,nmt}, \quad Q_{L,nmt}, \quad Q_{S,mt},$$

where D is deposits, M is mortgages, L is corporate loans, and S is securities/reserves.

Rates are:

$$R_{D,nmt}, \quad R_{M,nmt}, \quad R_{L,nmt}, \quad R_{S,t}.$$

Interpretation. Deposits are a funding product; mortgages and loans are asset products; securities/reserves are the safe liquid asset holdings created by reserve injections.

2.3 Data sources

- **Deposits.** FDIC Summary of Deposits gives bank-branch-level deposit volumes. Deposit rates come from RateWatch.
- **Mortgages.** HMDA provides mortgage volumes; mortgage rates are constructed at the bank-county-year level and matched to RateWatch.
- **Corporate loans.** Thomson Reuters syndicated loan data measure corporate loan issuance and rates.
- **Bank balance sheets.** Call Reports measure deposits, mortgages, commercial loans, and securities/reserves retained by banks.
- **Natural disasters.** Spatial Hazard Events and Losses Database for the United States (SHELDUS) provides property-loss shocks.
- **Reserve supply shocks.** Daily changes in the Treasury General Account, Reverse Repo Facility, and weekly total reserves help identify the reserve-spread response.

Interpretation. The data combine local product-market demand variation with bank-level balance-sheet information, which is necessary because reserves affect the entire balance sheet, not just one product.

2.4 Summary of key objects

- **Market-bank-year observations.** Deposit observations dominate the sample, with over 100,000 observations; mortgage and loan observations are smaller but still large.
- **Rates.** Mean rates are about 61.84 bps for deposits, 473.37 bps for mortgages, and 211.54 bps for corporate loans.
- **Volumes.** Mean volumes are about \$164 million for deposits, \$43 million for mortgages, and \$986 million for loans.
- **Illiquid assets.** Figure 1 shows the ratio of illiquid bank assets falling as injected reserves rise sharply after 2008 and again after 2020.

Interpretation. The summary statistics show that the products have very different rates, quantities, and market structures. This matters because reserves change all margins simultaneously.

2.5 Instruments

The paper uses three major sources of variation.

- **Natural-disaster instruments.** Property losses shift local demand for funds and internal reallocation across bank branches. These shocks help identify product demand and marginal costs.
- **Bartik deposit instrument.** Banks are exposed to different county-level deposit growth patterns through their branch networks. This generates bank-level variation in deposit-demand shocks.
- **TGA reserve-supply shock.** A rise in Treasury General Account balances removes reserves from the banking system, generating high-frequency variation in the reserve spread.

Interpretation. The central empirical challenge is simultaneity: banks choose quantities and rates together. The instruments generate exogenous shifts in demand or reserve supply that reveal slopes and cost interactions.

3 Models and empirical specifications

3.1 Graphical mechanism

A reserve injection forces the banking system to hold more liquid securities. If those securities raise the marginal cost of lending, the bank's marginal cost curve shifts left/up. The bank raises the loan rate from R_L to R'_L and loan quantity falls from Q_L to Q'_L .

$$\text{Reserve injection} \Rightarrow \text{higher marginal cost of lending} \Rightarrow R_L \uparrow, \quad Q_L \downarrow.$$

Interpretation. This is the core reserve supply channel: QE creates reserves, and reserves use balance-sheet capacity that would otherwise support lending.

3.2 Bank profit problem

Bank m chooses rates and security quantities to maximize the present value of profits:

$$\max_{R_D, R_M, R_L, Q_S} \sum_n Q_{L, nmt} (R_{L, nmt} - R_{L, t}^m) + \sum_n Q_{M, nmt} (R_{M, nmt} - R_{L, t}^m) \quad (1)$$

$$+ Q_{S, mt} (R_{S, t} - R_{S, t}^m) - \sum_n Q_{D, nmt} (R_{D, nmt} - R_{D, t}^m) - C(\Theta_{mt}). \quad (2)$$

Here $C(\Theta_{mt})$ is the balance-sheet cost function and

$$\Theta_{mt} = (Q_{D,nmt}, Q_{M,nmt}, Q_{L,nmt}, Q_{S,mt}).$$

Interpretation. The bank earns spreads on loans, mortgages, and securities, pays spreads on deposits, and incurs balance-sheet costs. The balance-sheet cost function is where reserves affect lending.

3.3 Demand system

Depositor j choosing bank m in market n and time t has utility

$$u_{D,jnmt} = \alpha_D R_{D,nmt} + X_{D,nmt} \beta_D + \delta_{D,nmt} + \varepsilon_{D,jnmt}. \quad (3)$$

The composite-good desirability of deposits is

$$\psi_{D,nt} = \log \left(\sum_m \exp(\alpha_D R_{D,nmt} + X_{D,nmt} \beta_D + \delta_{D,nmt}) \right). \quad (4)$$

Similar demand systems are used for mortgages and loans. For deposits, $\alpha_D > 0$ because depositors like higher rates; for mortgages and loans, rate coefficients are negative because borrowers dislike higher borrowing rates.

Interpretation. This modified logit framework estimates product demand without needing to directly observe a conventional outside good at the bank-market-year level.

3.4 Demand estimating equations

The demand system is estimated with two stages. First, product rates are instrumented:

$$R_{p,nmt} = a_{p,nt} + \nu_p z_{p,nmt} + X_{p,nmt} \gamma_p + e_{p,nmt}. \quad (5)$$

Then quantities or market shares are related to instrumented rates:

$$\log Q_{p,nmt} = \chi_{nmt} + \alpha_p R_{p,nmt} + X_{p,nmt} \beta_p + \xi_{p,nmt}. \quad (6)$$

Interpretation. The first stage isolates exogenous interest-rate variation. The second stage estimates how borrowers and depositors respond to that variation.

3.5 Outside-option estimation

For deposits and mortgages, the paper estimates the sensitivity of market-level quantities to the aggregate observed desirability parameter:

$$\psi_{p,nt}^o = \rho_{p,t} + \rho_{p,n} + \theta_p z_{p,nt} + X_{p,nt} \theta_p + \varepsilon_{p,nt}^o, \quad (7)$$

$$\log \bar{Q}_{p,nt} = \alpha_{p,t} + \alpha_{p,n} + \beta_{p,o} \psi_{p,nt}^o + X_{p,nt} \rho_p + \eta_{p,nt}. \quad (8)$$

Interpretation. This step translates bank-level market-share responses into market-level quantity elasticities, which are needed for the aggregate counterfactual.

3.6 Cost function and Hessian

The balance-sheet cost function is summarized locally by a Hessian matrix

$$H_{ij} = \frac{\partial^2 C(\Theta_{mt})}{\partial Q_i \partial Q_j}, \quad i, j \in \{D, ML, S\}. \quad (9)$$

The estimated Hessian is

$$H = \begin{pmatrix} 1.06 & -0.66 & -0.70 \\ -0.66 & 0.53 & 0.39 \\ -0.70 & 0.39 & 0.51 \end{pmatrix}.$$

Interpretation. The positive diagonal terms mean larger quantities raise own marginal costs. The positive $H_{S,ML} = 0.39$ means more securities/reserves raise the marginal cost of loans and mortgages. The negative $H_{S,D} = -0.70$ means securities/reserves lower the marginal cost of deposits.

3.7 TGA regression for the reserve spread

The high-frequency reserve-supply regression is

$$\Delta ReserveSpread_t = \alpha + \beta \Delta TGA_t + \gamma \Delta RRP_t + \theta \Delta TSY_w + FixedEffects + \varepsilon_t. \quad (10)$$

Interpretation. The TGA absorbs reserves from the banking system. A positive TGA shock is therefore equivalent to a negative shock to bank reserves for a fixed aggregate reserve supply.

3.8 Counterfactual reserve injection

For each year from 2008 to 2017, the authors inject the actual quantity of QE reserves and solve for the equilibrium set of rates and quantities. The equilibrium is a fixed point in which each bank adjusts deposit, mortgage, and loan rates while holding additional reserves.

$$\{R_D, R_M, R_L, Q_D, Q_M, Q_L, Q_S\}^{counterfactual} = \text{FixedPoint}(\text{demand system, cost function, reserve injection}).$$

Interpretation. The counterfactual isolates the effect of changing the reserve supply while preserving the demand and cost estimates from the structural model.

4 Identification and empirical design

4.1 What identifies demand slopes

- Deposit, mortgage, and loan demand slopes are estimated from market-bank-year variation in rates and quantities.
- Natural-disaster exposure shifts demand for local funds and allows rates to move for reasons not directly caused by local product demand in unaffected markets.
- The instrumented rate coefficient is positive for deposits and negative for mortgages and loans, matching economic intuition.

Interpretation. Demand slopes are crucial because the aggregate lending effect depends on how much loan quantities fall when loan rates rise.

4.2 What identifies the cost function

- The paper does not need reserve-supply shocks to estimate every cost interaction directly.
- Instead, it estimates how marginal costs respond to bank-level shocks in balance-sheet quantities.
- Symmetry of the Hessian is critical: if changes in loans affect the marginal cost of securities, then changes in securities affect the marginal cost of loans.
- A separate TGA regression identifies the securities-securities term H_{SS} .

Interpretation. This is the clever step. The authors use demand shocks to infer cost interactions and then use Hessian symmetry to learn how reserve injections affect loan costs.

4.3 Why corporate loans are the main outcome

- Deposit demand is relatively inelastic at the aggregate level.
- Mortgage demand responds less to rates than corporate loan demand.
- Corporate loan demand is much more elastic, so when reserve supply raises rates, loans absorb the biggest quantity decline.

Interpretation. The reserve supply channel is not mainly about deposit quantities falling. It is about reserves raising lending costs and crowding out the most elastic asset product: corporate loans.

4.4 Validation using IOER–FFR spread

The paper compares the model-implied reserve spread to the IOER–FFR spread observed in the data. The two series move similarly and have a correlation of about 0.50.

Interpretation. This is an external validation exercise. The reserve spread is not mechanically fitted by the counterfactual, so the similarity supports the model’s reserve-cost mechanism.

4.5 Main limitations

- The model is structural and depends on the validity of instruments and functional-form restrictions.
- The cost function is summarized locally by a Hessian, which may miss nonlinearities outside the observed range.
- Syndicated loan data may not represent all forms of bank credit.
- The counterfactual focuses on the reserve-supply side of QE, not all macroeconomic stimulus channels of QE.
- Bank regulation and balance-sheet constraints may change over time in ways not fully captured by the estimated cost function.

Interpretation. The conclusion is not that QE always reduces total welfare. It is that the reserve supply component of QE has a bank-lending crowd-out effect.

5 Tables and figures

5.1 Figure 1 and Figure 2: reserve supply and the mechanism

Read:

- **Figure 1** shows a large expansion of total reserves after 2008 and again after 2020.
- The fraction of illiquid assets on bank balance sheets falls as injected reserves rise.
- **Figure 2** illustrates the core mechanism: more reserves shift the bank's marginal cost curve for lending upward/left.
- The loan rate rises from R_L to R'_L and the loan quantity falls from Q_L to Q'_L .

Economic message. QE creates safe assets that banks must hold. These reserves use balance-sheet capacity and can displace lending to firms.

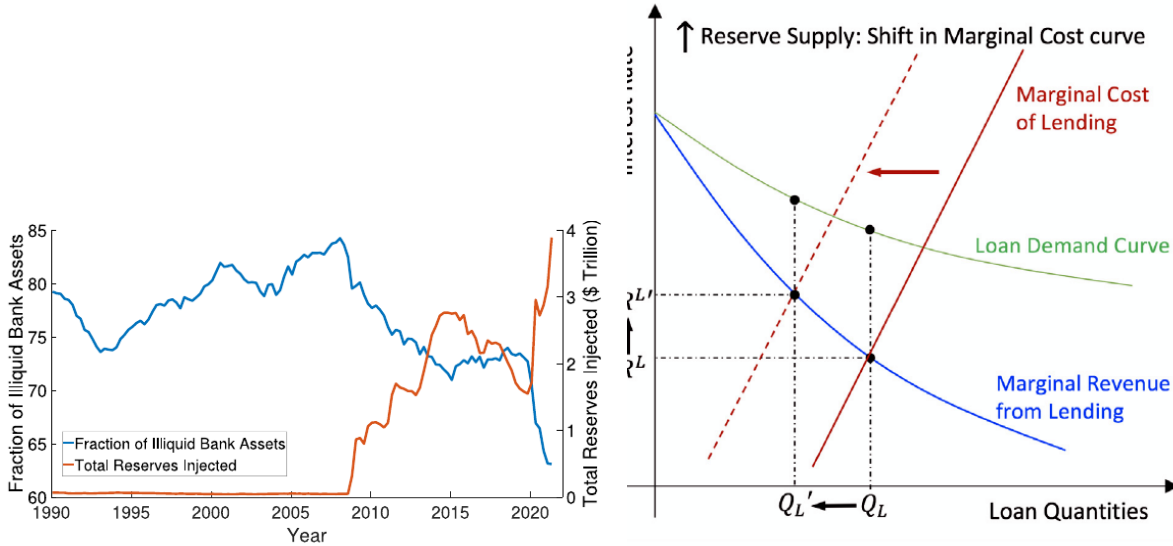


Fig. 1. Supply of Central Bank Reserves and Bank Asset Illiquidity.

This figure plots the total reserves of US depository institutions and their ratio

This figure illustrates the effect of an increase in reserves on the loan market. A

(a) Supply of reserves and bank asset illiquidity

(b) Loan-market mechanism of a reserve injection

5.2 Table 1, Table 2, and Table 3: summary statistics and demand systems

Read:

- **Table 1** shows the scale of product markets. The mean deposit rate is 61.84 bps, mortgage rate is 473.37 bps, and loan rate is 211.54 bps.
- **Table 2** reports strong first stages and economically intuitive 2SLS demand estimates: deposit market share rises with deposit rates, while mortgage and loan market shares fall with borrowing rates.
- The estimated loan market-share coefficient is -310.37 , and the mortgage coefficient is -533.93 .
- **Table 3** reports the outside-option estimates for deposits and mortgages. The 2SLS coefficient on aggregate desirability is about 0.05 for both deposit and mortgage quantities.

Economic message. Loan demand is much more rate sensitive than deposit or mortgage demand. This is why reserve-induced rate increases mainly show up as a decline in corporate loans.

Table 1
Summary statistics (Market-Bank-Year Level).
This table reports summary statistics of bank deposits, mortgages, and loans at the market-bank-year level. Rates are reported in basis points and volumes are in millions. For a given variable, #Obs refers to the total number of observations. The sample period is from 2001 to 2017.

	# Obs	Mean	25th Pct.	50th Pct.	75th Pct.	Std. Dev.
Log Deposit Market Share	105551	-3.00	-2.69	-4.07	-1.63	1.79
Deposit Volume	105551	163.96	63.60	29.09	144.44	874.39
Deposit Rate	105551	61.84	25.71	10.83	85.42	76.63
Log Mortgage Market Share	75470	-3.91	-3.57	-4.93	-2.62	1.78
Mortgage Volume	75470	42.85	7.44	2.41	23.29	266.77
Mortgage Rate	75470	473.37	472.73	350.00	586.25	124.20
Log Loan Market Share	23393	-4.98	-4.88	-6.46	-3.47	1.99
Loan Volume	23393	985.87	147.47	46.22	572.50	3227.84
Loan Rate	23393	211.54	197.27	132.30	275.00	114.98

(a) Table 1: summary statistics

Table 2
Demand system estimates.
This table reports the two-stage least squares results for estimating price disutility of deposit, mortgage, and loan demand systems:

$$R_{D,mt} = \alpha_{D,mt} + \gamma_D z_{D,mt} + X_{D,mt} \gamma_D + \epsilon_{D,mt},$$

$$\log Q_{D,mt} = \beta_D + \alpha_D R_{D,mt} + X_{D,mt} \beta_D + \epsilon_{D,mt}.$$

These regressions are run at the market-bank-year level. Loan loss provision is the ratio of loan loss provision over total loans, lag insured deposit ratio is the ratio of insured deposits over total liabilities lagged by 1 year, and log property damage is the county-level loss from natural disasters times the bank's branch number share in that county. For the deposit, mortgage and loan rates, 0.01 means 1%. The sample period is from 2001 to 2017. We report standard errors clustered by bank in the parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

Panel (a): First Stage panel regression			
	(1) Deposit rate	(2) Mortgage rate	(3) Loan rate
IV	1.69*** (0.26)	12.17*** (2.90)	3.18*** (0.57)
Loan Loss Provision	0.01*** (0.00)	-0.02 (0.01)	0.05* (0.03)
Lag Insured Deposit Ratio	0.00*** (0.00)		
Log Property Damage	-0.00*** (0.00)	-0.00** (0.00)	
R ²	0.87	0.91	0.29
Adj. R ²	0.80	0.85	0.27
Num. of Obs.	105551	75470	23393
Market-Year F.E.	Y	Y	Y
Panel (b): 2SLS Panel regression			
	(1) Deposit market share	(2) Mortgage market share	(3) Loan market share
Rate (with IV)	151.32*** (49.07)	-533.93*** (167.24)	-310.37*** (89.25)
Loan Loss Provision	-1.41 (1.26)	-14.53 (9.98)	19.56 (12.27)
Lag Insured Deposit Ratio	-1.97*** (0.18)		
Log Property Damage	1.26*** (0.02)	0.81*** (0.07)	
Num. of Obs.	105551	75470	23393
Market-Year F.E.	Y	Y	Y

(b) Table 2: demand-system estimates

Table 3
Outside option estimates (Deposits and Mortgages).

This table reports two-stage least squares results for estimating the sensitivity of market-level quantities to the aggregate observed desirability parameter ψ_{mt}^o :

$$\psi_{D,mt}^o = \rho_{D,I} + \rho_{D,mt} + \theta_D z_{D,mt} + X_{D,mt} \theta_D + \epsilon_{D,mt}^o,$$

$$\log Q_{D,mt} = \alpha_{D,I} + \alpha_{D,mt} + \beta_{D,mt} \psi_{D,mt}^o + X_{D,mt} \rho_D + \eta_{D,mt}.$$

The regression is run at the market-year level. We include market-year level controls, including average age and income of the population, fraction of residents college degree, log population, annual house price growth, and log property losses due to natural disasters. For the aggregate desirability of deposits and mortgages, 0.01 means 1%. The sample period is 2001–2017. We report standard errors clustered by county in the parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

Panel (a): First Stage		
	(1) Deposit $\psi_{D,mt}^o$	(2) Mortgage $\psi_{M,mt}^o$
IV	0.02*** (0.01)	-0.45*** (0.13)
R ²	0.99	0.94
Adj. R ²	0.99	0.94
Within Adj. R ²	0.98	0.72
Num. of Obs.	43255	22447
Controls	Y	Y
Year F.E.	Y	Y
County F.E.	Y	Y

Panel (b): 2SLS		
	(1) Deposit quantity	(2) Mortgage quantity
ψ^o (with IV)	0.05 (0.06)	0.05 (0.03)
Num. of Obs.	43255	22447
Controls	Y	Y
Year F.E.	Y	Y
County F.E.	Y	Y

5.3 Tables 4–6: cost function and reserve-spread estimates

Read:

- **Table 4** estimates how marginal costs and quantities respond to natural-disaster and Bartik deposit shocks.
- Under the natural-disaster instrument, loan/mortgage cost rises by 2.46 bps and deposit cost falls by 3.17 bps.
- Under the Bartik deposit shock, deposit cost rises by 30.98 bps, while mortgage/loan cost falls by 24.57 bps.

- **Table 5** estimates that a \$1 trillion increase in the TGA balance lowers the reserve spread by 7.88 bps, consistent with reserves being removed from the banking system.
- **Table 6** gives the implied Hessian. Securities/reserves raise loan and mortgage marginal costs ($H_{S,ML} = 0.39$) and lower deposit marginal costs ($H_{S,D} = -0.70$).

Economic message. The cost estimates are the empirical backbone of the reserve supply channel. More reserves make bank lending more expensive, not cheaper.

Table 4
Cost function regression estimates.
This table reports the sensitivity of bank-level costs and quantities to losses from natural disasters and a Bartik deposit shock as in Eqs. (23) and (24). Sheldus Instrument refers to property losses due to natural disasters as explained in Section 4.3. Bartik Deposit Instrument refers to a Bartik-style instrument of deposit growth as explained in Section 4.3. Rates are in basis points and quantities are in millions. The sample period is from 2001 to 2017. We report standard errors clustered by bank in the parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

Panel (a): Results using natural disaster instrument					
	Deposit cost (1)	Mtg/Loan cost (2)	Deposit Vol (3)	Mtg/Loan Vol (4)	Security Vol (5)
Sheldus Instrument	-3.17*** (0.32)	2.46*** (0.92)	28.31*** (8.11)	26.37*** (6.25)	18.97** (8.12)
Loan Loss Provision	-5.96* (3.26)	1.91 (1.34)	6.98 (7.79)	527.36*** (202.27)	1.51 (2.40)
R ²	0.59	0.82	0.00	0.01	0.00
Adj. R ²	0.59	0.82	0.00	0.01	0.00
Num. of Obs.	53651	12733	118942	119236	118923

Panel (b): Results using Bartik deposit shock					
	Deposit cost (1)	Mtg/Loan cost (2)	Deposit Vol (3)	Mtg/Loan Vol (4)	Security Vol (5)
Bartik Instrument	30.98*** (7.96)	-24.57* (14.85)	1048.26** (417.94)	474.34*** (136.09)	1084.52** (508.92)
Loan Loss Provision	-4.67 (3.28)	-7.80** (3.47)	7.19 (48.79)	153.96 (93.89)	-48.37 (48.10)
R ²	0.44	0.82	0.00	0.01	0.00
Adj. R ²	0.44	0.82	0.00	0.00	0.00
Num. of Obs.	50091	12093	62352	62458	62346

TGA regression estimates.

This table reports the results from the TGA regression (28). Rates are in basis points and quantities are in trillions. The sample period is from 2001 to 2017. *, **, and *** denote significance at the 10%, 5%, and 1% level, respectively.

Change in TGA Balance	-7.88*** (1.82)
Change in RRP	0.03*** (0.00)
Change in Fed Sum	0.00 (0.00)
R ²	0.46
Adj. R ²	0.46
Num. of Obs.	1770

Table 6

Implied Hessian.

This table reports the implied Hessian matrix H based on the cost function estimates. Please refer to Section 4 for a detailed description of the estimation. The Hessian matrix reports the impact of an extra \$1 million dollars per branch of a balance sheet quantity on the number of basis points by which a bank's marginal cost changes.

	$\frac{\partial C}{\partial Q_D}$	$\frac{\partial C}{\partial Q_{L+}}$	$\frac{\partial C}{\partial Q_S}$
Q_D	1.06	-0.66	-0.70
Q_{ML}	-0.66	0.53	0.39
Q_S	-0.70	0.39	0.51

(a) Table 5: TGA regression estimates

(b) Table 6: implied Hessian matrix

5.4 Table 7 and Figures 3–5: counterfactual QE results

Read:

- **Table 7** shows that QE reserve injections raise deposit rates by 15.56 bps, mortgage rates by 8.38 bps, loan rates by 7.37 bps, and securities yields by 11.55 bps.
- Quantities change by +\$0.0809 trillion for deposits, -\$0.0063 trillion for mortgages, -\$0.1342 trillion for loans, and +\$1.8323 trillion for securities.

- **Figure 3** shows that counterfactual syndicated lending without reserve injections would have been higher than observed lending after 2011.
- **Figure 4** shows reserve injections alongside the estimated loan issuance crowded out.
- **Figure 5** compares the observed IOER–FFR spread with the model-implied reserve spread; the two series track each other reasonably well.

Economic message. The model turns estimated demand and cost curves into a quantitative QE counterfactual: the reserve supply component of QE crowds out a meaningful amount of corporate bank lending.

Table 7

Counterfactual results: QE

This table reports the results of our counterfactual analysis that injects the actual amount of reserves QE supplied for each year from 2008 to 2017. We compute the effects on rates and quantities, and report the average across years.

Average change in rates (in basis points)				Average change in quantities (in Trn dollars)			
Deposits	Mortgages	Loans	Securities	Deposits	Mortgages	Loans	Securities
15.56	8.38	7.37	11.55	0.0809	−0.0063	−0.1342	1.8323

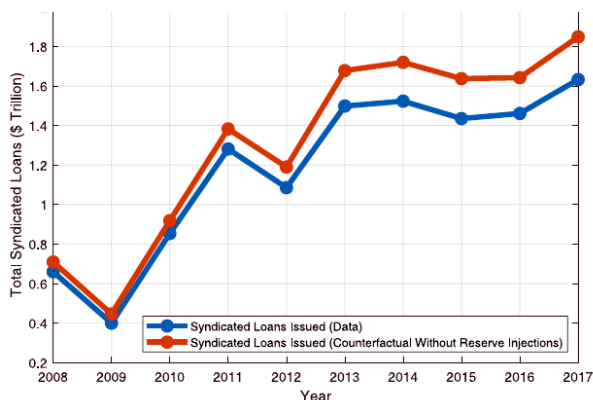


Fig. 3. Observed and Counterfactual Lending Quantities.

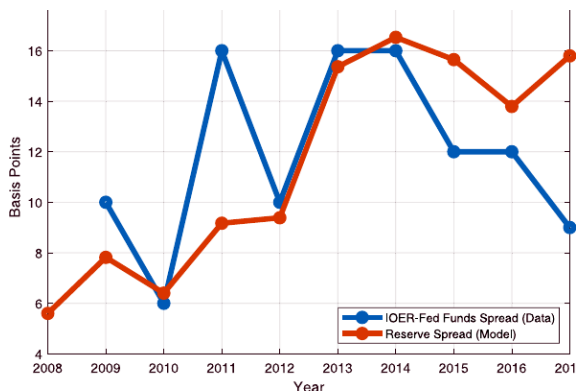


Fig. 5. IOER-Fed Funds Spread and Model-Implied Reserve Spread.

(a) Figure 3: observed and counterfactual syndicated loans

(b) Figure 5: IOER–FFR and model-implied reserve spread

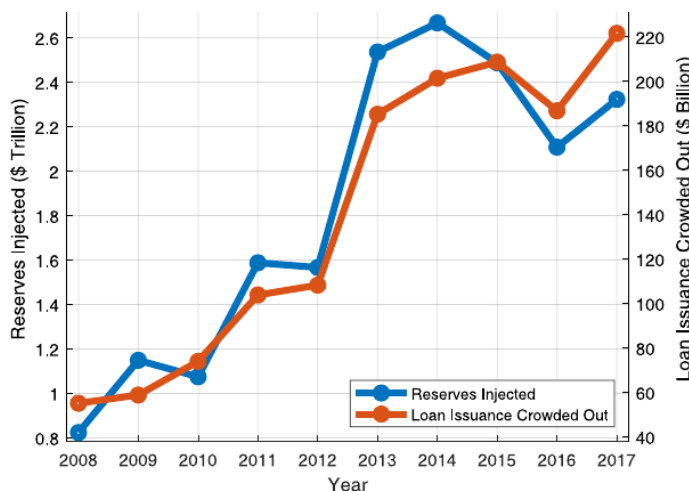


Fig. 4. Reserve Supply and Reduction in Corporate Loan Issuance.

6 Short summary

- The paper identifies a reserve supply channel of QE.
- QE buys assets and pays with reserves. Those reserves must be held inside the banking system.
- Banks face balance-sheet costs that depend on deposits, mortgages, loans, and securities/reserves.
- The estimated cost function implies that additional reserves raise the marginal cost of providing loans and mortgages.
- The estimated demand system implies that corporate loan demand is relatively elastic.
- As a result, reserve injections mostly crowd out corporate loans rather than deposits or mortgages.
- The counterfactual QE exercise finds that reserves raise loan rates by 7.37 bps and reduce loans by about \$134.2 billion per year.
- Each dollar of QE reserves reduces bank lending by about 7.67 cents.
- The model-implied reserve spread matches the broad magnitude and dynamics of the IOER–FFR spread.
- The results do not claim that all QE channels are contractionary. They isolate one reserve-supply component that works against bank lending.

7 Conclusion

7.1 Core conclusion

Diamond, Jiang, and Ma show that reserve injections from QE can crowd out bank lending. The crowding out arises because reserves use balance-sheet capacity and raise the marginal cost of supplying loans and mortgages. In the authors’ counterfactual, the reserves created by QE from 2008–2017 reduce corporate bank lending by about \$134 billion per year.

7.2 Economic intuition

The intuition is that reserves are safe, liquid assets that banks are required to hold after central bank asset purchases. If balance-sheet space is costly, then more reserves make it more expensive for banks to hold other assets, especially loans. Banks respond by raising loan rates. Because corporate loan demand is relatively elastic, loan quantities fall.

7.3 Takeaway

The paper adds a banking-sector balance-sheet channel to the analysis of QE. Asset purchases may stimulate the economy through duration, signaling, or portfolio-balance effects, but the reserve supply created by those purchases can reduce lending capacity inside the banking system. The net effect of QE therefore depends not only on what assets the central bank buys, but also on the cost of absorbing the reserves it creates.